

Supplementary Material

Task-Oriented Semantic Granularity Selection for Bandwidth-Constrained V2V Cooperative Perception

TABLE I

Ego-side online task cues used by the CA-TOSG selector ($d = 21$). All cues are computed locally on the ego vehicle before message selection, removing the causality issue that would arise from depending on the collaborator’s state.

Category	Cue	Definition
Agent context	num_cavs	number of CAVs in scene
	ego_num_objects	ego-detected objects
Frame shape	ego_origin_lidar_shape_0/1	raw LiDAR tensor shape
Range stats	pcd_num_points	total point count
	pcd_mean_range	mean LiDAR range
	pcd_max_range	max LiDAR range
	pcd_std_range	range std. dev.
Distance bins	pcd_near_20m	points within 20 m
	pcd_mid_20_50m	points 20–50 m
	pcd_far_50_80m	points 50–80 m
	pcd_very_far_80m	points beyond 80 m
Quadrant cnt.	pcd_front_points	front-half points
	pcd_back_points	rear-half points
	pcd_left_points	left-half points
	pcd_right_points	right-half points
Long-range	pcd_front_far_30m	front points > 30 m
	pcd_front_far_50m	front points > 50 m
Density bins	pcd_density_0_20	density 0–20 m
	pcd_density_20_50	density 20–50 m
	pcd_density_50_80	density 50–80 m

This document carries material moved out of the main paper for length. It is not a second set of results: the two appendices below are the prior-protocol arms labelled as such in the main text, and the cue table is the definition list of the 21 committed ego-side cues. Section and label names are preserved so that the main text’s pointers resolve against this document.

I. Ego-Side Online Task Cues: full definition list

The selector’s task-cue vector $z_t \in \mathbb{R}^{21}$ is defined in full below; the main paper lists five representative cues and points here.

II. When Are Perception Cues Necessary? LDPC Cliff versus JSCC Graceful Degradation

Protocol note. A prior-protocol arm, not re-evaluated under the frozen P2 protocol: its selector is trained and evaluated inside the codec-comparison pipeline, sharing the retired engine’s BLER lookup and 200-realisation CSI convention. Its numbers may not appear in the same sentence as a frozen-replay number, and no main-text claim rests on them.

Under LDPC + QAM a one-line threshold tracks the learned selector (Section ??); does that equivalence survive a feature codec without a sharp SNR cliff? We replace the feature branch with importance-map JSCC [1] and re-run the per-frame pipeline: per-frame JSCC F1 is re-cached at each SNR (1,000 frames per SNR, AWGN and Rayleigh), the $\{L, \text{Feature-JSCC}\}$ oracle relabelled and the selector retrained—no perception-backbone change.

The mechanism is a codec property. The LDPC + QAM feature F1 cliffs over the 8–12 dB decoding transition (onset 8.0 dB), so SNR is highly informative; the learned JSCC feature F1 is essentially flat at ≈ 0.89 across 0–20 dB, so the estimated SNR says almost nothing about whether a feature message will help on a given frame.

This flips the comparison (Table II). Both codecs are evaluated on the same 2,170 test frames under the identical 200-seed protocol, separating (a) whether the graceful-channel gain exists and resides in the cues—estimated leakage-free in-distribution by 5-fold cross-validation, every frame scored by a selector that never saw it and τ tuned on the training folds only—from (b) whether a deployed, validate-frozen selector realises it, under the headline cross-split protocol.

(a) In-distribution. Under LDPC + QAM the AWGN edge over the best SNR threshold is only +0.002 F1 (95% CI [+0.001, +0.003]): small, and—as the feature ablation shows—not attributable to the perception cues alone (Section ??).¹ Under JSCC the best SNR threshold collapses to the Fixed- L operating point—it cannot gate on an uninformative SNR—and the cue-based edge, measured by the in-distribution k -fold estimator, widens to +0.022 F1 ([+0.020, +0.025]) under AWGN, +0.018 ([+0.016, +0.020]) under Rayleigh, and +0.020 ([+0.018, +0.022]) under OFDM on the test split, and to +0.012–+0.015 on the Culver-City domain shift (all CIs exclude zero). In the separate 200-realisation held-out comparison on validate frames, the cue-based selector recovers 56–62% of the clairvoyant oracle’s headroom over

¹This in-distribution per-frame edge (+0.002) is distinct from two other LDPC-regime quantities: the feature-ablation gain of the cues (Full vs. channel-state-only, +0.0090, 95% CI [+0.0089, +0.0091], Section ??) and the matched-payload margin over the retuned threshold (+0.0032, Section ??). The three measure the selector along the cue, the equal-bandwidth, and the per-frame axis respectively. Under the corrected single-collaborator convention all three are positive, with the ordering $+0.002 < +0.0032 < +0.0090$: the cue axis, smallest and indistinguishable from zero under the retired full-collaborator accounting (−0.0002), is now the largest of the three — weakening the single message widened the gap the cues have to close, and they close part of it.

TABLE II

Learned-selector edge over the best SNR-threshold rule on the same 2,170 test frames (200-seed protocol), reported two ways. In-dist. edge: leakage-free 5-fold cross-validation on the evaluation split (each frame scored by a selector that never saw it, τ tuned on the training folds only). Deployed edge: selector trained on validate and frozen, applied cross-split (the headline protocol). In-distribution the JSCC edge is 9–11 $\times$ the LDPC-cliff edge—an SNR threshold cannot reach it—while under LDPC the margin is the selector’s function form, not a cue gain (cues add no significant F1, Section ??). The deployed JSCC edge is negative: the validate-trained selector does not transfer the graceful-channel gain across the domain shift (Appendix II(b)).

Feature codec	Channel	In-dist. edge	Deployed edge
LDPC + QAM (cliff)	AWGN	+0.002 [+0.001, +0.003]	−0.002 [−0.003, −0.001]
Importance-map JSCC	AWGN	+0.022 [+0.020, +0.025]	−0.004 [−0.007, −0.001]
Importance-map JSCC	Rayleigh	+0.018 [+0.016, +0.020]	−0.008 [−0.011, −0.006]
Importance-map JSCC	OFDM	+0.020 [+0.018, +0.022]	−0.005 [−0.008, −0.003]

Fixed L across the three channels. The two quantities are distinct and are reported separately: under AWGN the oracle headroom is 0.0291 F1 and the selector recovers +0.0181 of it, with 0.0275 and +0.0158 under Rayleigh and 0.0281 and +0.0158 under OFDM. In-distribution, then, the graceful-channel decision is content-bound and carried by the ego-side cues—a gain no SNR threshold reaches on these frames, and one that (b) below shows does not survive the cross-split move.

(b) Cross-split deployment. The signal does not transfer: deployed on test the frozen selector over-selects the feature action (JSCC C-request rate 0.14 $\rightarrow$ 0.42) and falls just below Fixed- L —edge −0.004 (AWGN), −0.008 (Rayleigh), −0.005 (OFDM) on test, ≈ 0 on Culver-City. The gain is present but not portable; capturing it under cross-domain deployment, e.g. by online or self-supervised adaptation [2], is future work.

Whether task-oriented cues are needed is determined by the feature codec’s channel response. Under a cliff codec the decision is channel-bound and SNR suffices; under a graceful codec it becomes content-bound and the ego-side cues—which predict whether object-level detection is already adequate—become the operative signal, in-distribution. Closing the cross-split transfer gap (b) is the remaining challenge.

III. Boundary of the Method: a Second Detection Backbone

Convention. This arm was trained and evaluated on the full-collaborator caches, i.e. before the corrigendum of Change-log P0: it is internally consistent (its own budget, its own N_{cw} , its own references) and its relative conclusion stands, but its absolute F1 and payload figures are on a different accounting basis from the single-collaborator main experiment and are not tabulated beside it.

Positioning. This appendix reports a limitation, not a generality result. It is placed here because its conclusion is negative: the selection procedure developed in the main text is in-sample effective on a second backbone but does not transfer to held-out splits under the equal-budget protocol. No claim in the main text depends on it.

TABLE III

Second backbone (SECOND), equal-budget controlled. Paired $\Delta F1$ of the frozen selector against the SNR-threshold rule tuned to the same budget, over the same 200 CSI samplings; 95% frame-level paired bootstrap intervals. Positive favours CA-TOSG. Second-backbone arm, not deployed; descriptive, no decision is taken from this table.

Split	$B_{\max} = 0.10$	$B_{\max} = 0.20$	$B_{\max} = 0.30$
Validate (in-sample)	+0.0058 [+0.0057, +0.0059]	+0.0095 [+0.0094, +0.0096]	+0.0081 [+0.0081, +0.0082]
OPV2V test	−0.0014 [−0.0015, −0.0013]	−0.0047 [−0.0048, −0.0046]	−0.0066 [−0.0067, −0.0065]
Culver-City	−0.0029 [−0.0030, −0.0028]	−0.0117 [−0.0119, −0.0115]	−0.0167 [−0.0169, −0.0165]

Every result above uses one perception backbone. To test whether the selection procedure—not the particular features—is what carries the behaviour, we repeat the whole pipeline on a second detector, SECOND [3], under an equal-budget controlled comparison: the feature message is charged the same per-frame channel budget as the mainline ($B_F \equiv 0.99$ Msym, the same N_{cw} and the same BLER tables), so the two backbones are compared at equal channel cost. We do not claim that SECOND’s feature tensor compresses to that size, and we declare no codec for it.²

Both SECOND checkpoints are the official OpenCOOD ones, and we verify them before use rather than trusting the file: after a lossless axis-order conversion required by the spconv version in our environment, official inference reproduces the published AP@0.7 to +0.0002 on both splits for the late-fusion model (0.7752 vs. 0.775; 0.6822 vs. 0.682) and to within +0.0019 for the intermediate model. The three per-frame branches (E , L , F) are then re-derived with the same scorer, the same canonical union ground truth and the same IoU-0.5 convention as the mainline, and the entire downstream pipeline—grid expansion, scene-level 9-fold LOSO, the budget walk, and the 200-realisation paired replay—is re-run with the mainline code, each stage first verified to reproduce its own committed mainline product bit-for-bit.

The result is that CA-TOSG is in-sample effective but does not transfer under the equal-budget protocol. Table III gives the paired comparison against an SNR-threshold rule tuned on validate for the same target budget. On validate—the split the selector is trained on—it is ahead at every budget (+0.0058 to +0.0095 F1). On both held-out splits it is behind at every budget, with all confidence intervals excluding zero.

Two mechanisms account for the gap, and both are visible in the arm’s own numbers. First, the L – F margin is narrower on this backbone: SECOND’s object-level branch scores 0.869 / 0.879 / 0.879 mean per-frame F1 on validate / test / Culver-City against a compressed-feature branch

²The transmitted tensor was measured, not assumed: SECOND’s cross-link BEV tensor is 6,758,400 elements per collaborator before compression and 352,000 at the auto-encoder bottleneck. Both are recorded as measurements; neither is used as the operative payload under the equal-budget convention.

at 0.897 / 0.904 / 0.939, so there is simply less to gain from choosing correctly, and the same selection error costs proportionally more. Second, the budget walk selects a different candidate family: all three frozen SECOND selectors carry `class_weight=balanced`, whereas all three mainline selectors carry `class_weight=None`. The reweighted family fits the in-sample class mix well—agreement with the oracle is 0.833 on validate—but transfers poorly, falling to 0.578 on test and 0.534 on Culver-City, where the selector degenerates to near-always- L (E recall 0.027 on test and 0.000 on Culver-City). The E -collapse of Section ?? is therefore worse on this backbone, not milder: under Rayleigh the selector’s ρ_E is 0.002 (test) and 0.000 (Culver-City) while the oracle spends 0.157 and 0.133. The large payload reductions this arm shows (67–90%) follow from that collapse to L and are not evidence of good selection.

We report this as a boundary of the present method rather than a tuning target: nothing was retrained and no threshold was adjusted after seeing these numbers.

References

- [1] Y. Sheng, H. Ye, L. Liang, S. Jin, and G. Y. Li, “Semantic communication for cooperative perception based on importance map,” *Journal of the Franklin Institute*, vol. 361, no. 6, p. 106739, 2024.
- [2] C. Liu, J. Chen, Y. Chen, R. Payton, M. Riley, and S.-H. Yang, “Self-supervised adaptive weighting for cooperative perception in V2V communications,” *IEEE Trans. Intell. Veh.*, vol. 9, no. 2, pp. 3569–3580, 2024, doi: 10.1109/TIV.2023.3345035.
- [3] Y. Yan, Y. Mao, and B. Li, “SECOND: Sparsely embedded convolutional detection,” *Sensors*, vol. 18, no. 10, p. 3337, 2018.